Printed Pages: 04 **Sub Code: EEE402/ NEE402**

Paper Id: 120243 **Roll No.** | | | | | | | | | | |

B TECH
(SEM IV) THEORY EXAMINATION 2017-18
Network Analysis and Synthesis

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

a. Write down the properties of a tree in the graph.

b. If $Z_L=4+2j$, what should be the value of Z_{th} for maximum power transfer.

c. Write down the limitation of Superposition Theorem.

d. If a network contains B branches, and N nodes, then the number of mesh current equations should be_____.

e. What are transients and why are they produced.

f. Determine the time constant of series RL circuit in which a constant supply of 100V is connected across the series combination of resistance of 4K Ω and inductance of 0.01mH

g. Define complex frequency.

h. Define Driving point function.

i. Write down the properties of Hurwitz polynomial.

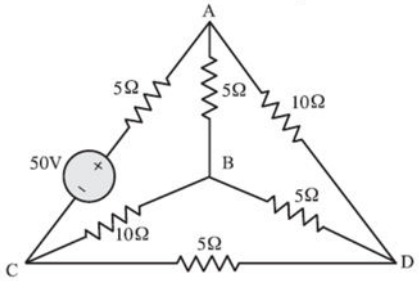
j. What are the limitations of Foster forms?

SECTION B

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2. Attempt any *three* of the following: **10 x 3 = 30**

a. (i) Find out the number of possible trees for the figure given below:



(ii) Obtain the oriented from the given Cut-set matrix:

$$[Q] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

b. Find the expression of the current i(t) in the circuit of figure 8 assuming that the switch is opened at t=0, the steady state having already reached before that.

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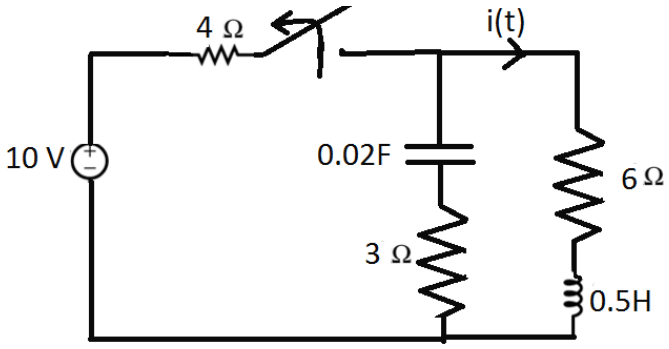
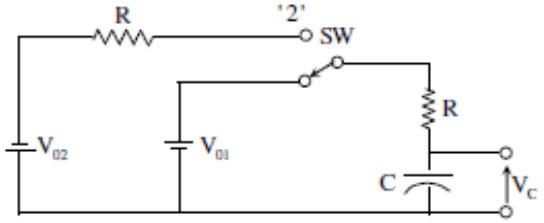
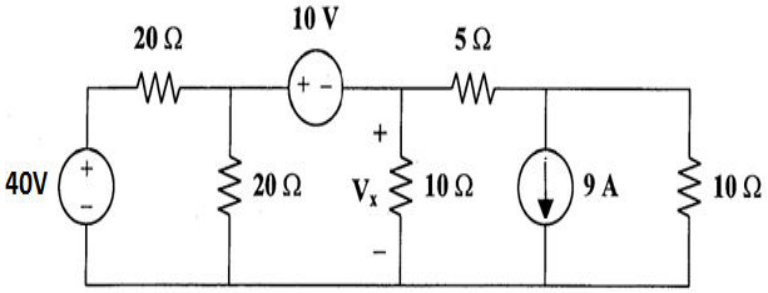


Figure 8

c. The switch SW shown in the circuit is kept at position '1' for a long duration. At t = 0+, the switch is moved to position '2'. Assuming $V_{02}<V_{01}$, find the voltage $V_C(t)$ across capacitor in the figure given below:



d. Explain the concept of duality. Find the dual of the network shown in figure1.



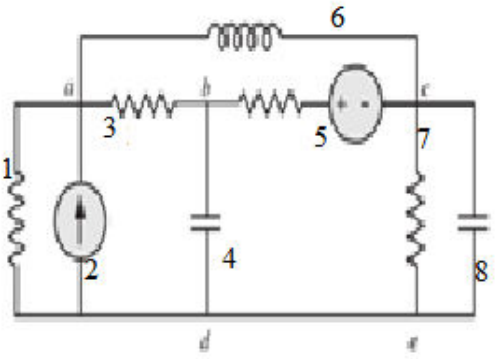
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e. Define Stermtest. Why it is used? Explain the process involved in Sterm test.

SECTION C

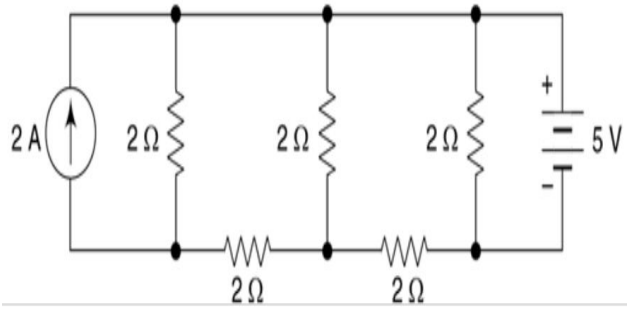
3. Attempt any *one* part of the following: 10 x 1 = 10

(a) Write down the Cut-set and Tie-set matrix for the circuit given in fig. Consider 3, 5 and 4 as twigs. Also write down its Impedance Matrix.



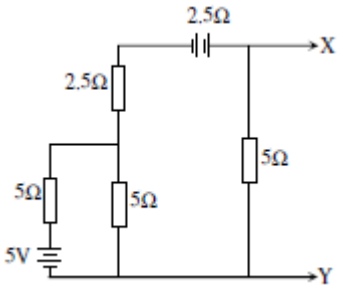
(b)For the network shown in the figure below, draw the oriented graph and create the tie-set matrix. Use this matrix to calculate current flowing through 5V source.

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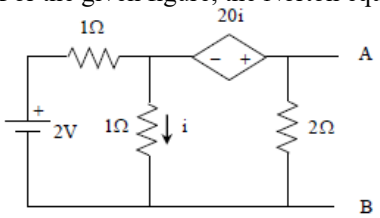


4. Attempt any *one* part of the following: 10 x 1 = 10

(a) Obtain the Thevenin's equivalent circuit across the terminals X and Y in the ccircuit shown elow:



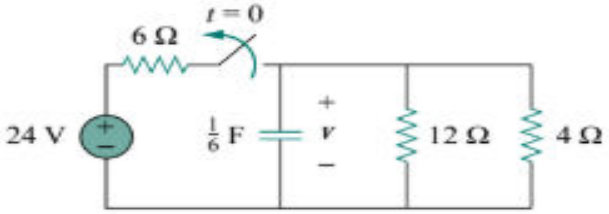
(b) For the given figure, the Norton equivalent circuit is to be determined across terminal AB.



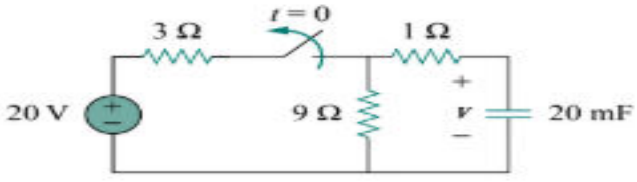
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5. Attempt any *one* part of the following: 10 x 1 = 10

(a) For the given circuit, find v(0+), dv(0+)/dt, $d^2v(0+)/dt^2$& v(∞).



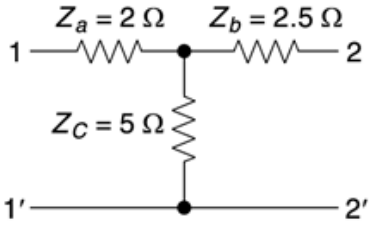
(b) Find the expression for capacitive voltage for t>0 and initial energy which is stored in the capacitor.



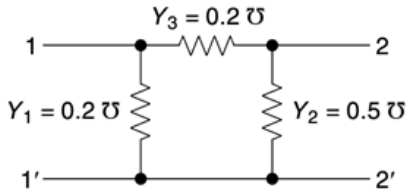
6. Attempt any *one* part of the following: 10 x 1 = 10

(a) Find the equivalent π-network for the *T*-network shown in the Fig.

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(b) Find the equivalent *T* -network for the *π*-network shown in the Fig.



7. Attempt any *one* part of the following: 10 x 1 = 10

(a) How the function can be identified as a positive real function? Explain with a suitable example.

(b) Differentiate between Foster and Cauer forms and also write down the steps to obtain Foster-I and Foster-II forms.

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